

SSN College of Engineering, Kalavakkam
Department of Computer Science and Engineering

Academic Year: 2023-2024 ODD

Batch: 2021-2025

III Year - V Semester

16th August, 2023

Common Courses Committee Meeting

Course: **UCS2503–Software Engineering**

Members Present:

Faculty: **Dr. A.Chamundeswari, Dr.Y.V.Lokeswari**

Student Representatives:

V Sem CSE A:

215001001 **Aathish S**
215001025 **Gayathri N**

V Sem CSE B:

215001066 **Pooja Premnath**
215001092 **Sanjai Balajee K G**

The meeting of common courses committee was held on 16th August 2023 for the course **UC S2503 – Software Engineering** under SSN (autonomous) 2021 regulations. The course plan, course outcomes and the assessment details were discussed.

Course Code	UCS2503	Course Name	Software Engineering					
Course Type	Theory	Course Category	Professional Core (PC)	L	T	P	E	C
				3	0	0	0	3
Regulation	2021		Academic Year	2023-2024 ODD				
Degree and Branch	B.E. Computer Science & Engineering		Batch	2021-2025				
Semester		V	Faculty Name	Dr. A. Chamundeswari & Dr. Y. V. Lokeswari				
Department Offering the Course			Computer Science and Engineering					

Course Outcomes

Upon completion of this course, the students should be able to:

1. Choose an appropriate process model and estimate project cost and effort required by applying software engineering principles(K5)
2. Identify and analyze the requirements and construct their models (K5)
3. Apply systematic procedure for software design (K3)
4. Compare and contrast the various testing and maintenance activities (K2)
5. Make use of agile development and Devops (K3)
6. Identify unethical issues and apply ethical practices for a given case study (K3)

Assessment Format and Details

Assessment Tool	Unit	CO1	CO2	CO3	CO4	CO5	CO6
CAT-1	I, II & III	Y	Y	Y			
CAT-2	III, IV & V			Y	Y	Y	
Assignment KL (K5)	I & II	Y	Y				Y
Assignment KL (K5)	III & IV			Y	Y		Y

CAT 1 assesses CO1, CO2 and CO3 (from Units I, II and III) mapping to PO1, PO2, PO3, PO5, PO6, PO7, PO9, PO10, PO11, PSO1, and PSO2.

CAT 2 assesses CO3, CO4 and CO5 (from Units III, IV and V) mapping to PO1 PO2, PO3, PSO1, and PSO2.

Assignment-1 involving Engineering knowledge, problem analysis, sustainability, ethics and life-long learning CO1, CO2 & CO6 (from units I and II) mapping to PO1, PO2, PO3, PO5, PO6, PO7, PO8, PO9, PO10, PO11, PO12, PSO1 and PSO2.

Assignment-2 involving Engineering knowledge, designing solutions, testing tools, Sustainability, ethics and life-long learning CO3, CO4 & CO6 (from units I, II, III, IV) mapping to PO1, PO2, PO3, PO8, PO10, PO12, PSO1 and PSO2.

Assessment				
Assessment I (100 Marks)		Assessment II (100 Marks)		End Semester Examination
Assignment	Written Test	Assignment	Written Test	Written Test
40 Marks	60 Marks	40 Marks	60 Marks	100 Marks
20% Marks		20% Marks		60% Marks

- Average of two Continuous Assessment Tests with a weightage of 30% as per college schedule
- Assignments with a weightage of 10% as per the course instructor's schedule.
- End Semester Examination with a weightage of 60% as per final examination schedule

Faculty

Dr.A.Chamundeswari

Dr. Y. V. Lokeswari

HoD-CSE

Dr. T. T. Mirnalinee

Date: 16-Aug-2023

Course Code	UCS2503	Course Name	Software Engineering					
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Semester		V	Faculty Name	Dr. A. Chamundeswari & Dr. Y. V. Lokeswari				
Department Offering the Course			Computer Science and Engineering					

COURSE OBJECTIVES

1. To understand the phases in a software project, estimate cost and effort.
2. To understand fundamental concepts of requirements engineering and Analysis Modeling.
3. To understand the various software design methodologies
4. To learn various testing techniques and maintenance measures.
5. To understand agile development and Devops.

UNIT I SOFTWARE PROCESS AND PLANNING

9

Introduction to Software Engineering; Objectives, Principles and Practices; The Software Development Life Cycle : Pre-development phases of the SDLC – Development specific phases of the SDLC – Post-development phases of the SDLC; Methodologies Paradigm and Practices : Process methodologies – Development paradigms – Development practices; Project Planning Process; Software Project Estimation: Decomposition techniques – Empirical estimation models – The make/buy decision – Project scheduling; Risk Management; Handling Ethical Dilemmas.

UNIT II REQUIREMENTS ANALYSIS AND SPECIFICATION

9

Software Requirements: Functional and non-functional – Security requirements – User requirements- System requirements – Software requirements document; Requirement Engineering Process: Feasibility studies – Requirements elicitation and analysis – Requirements validation – Requirements Management; Classical Analysis: Structured system analysis; Requirement modelling tools.

UNIT III SOFTWARE DESIGN

9

Design Concepts: Design process – Design concepts – Modularity, Coupling and cohesion – Design model – Modeling principles; Structured Design; Architectural Design: Architectural styles; Architecture for Network based Applications – Decentralized Architectures.

UNIT IV SOFTWARE TESTING

9

Software Testing Fundamentals; Internal and External Views of Testing: White box testing – Basis path testing – Control structure testing – Black box testing – Unit testing – Integration testing – Regression

testing – Validation testing – System testing – Security testing; Testing Tool; Debugging; So
Implementation: Coding Practices and Principles; Maintenance: Types.

UNIT V AGILE DEVELOPMENT AND DEVOPS

9

Agile Development: Agile Teams – Team and Scrum – Branches – Pull Requests – Reviews –
Integration – Agile Iterations – Reporting and fixing bugs; Dev/Ops: From development to deployment –
Three-Tier – Responsiveness, Service level objectives, and Apex – Releases and feature flags –
Monitoring and finding bottlenecks – Improving rendering and database performance with caching;
Security: Defending customer data in application.

TOTAL PERIODS: 45

COURSE OUTCOMES

On successful completion of this course, the student will be able to:

1. Choose an appropriate process model and estimate project cost and effort required by applying software engineering principles(K5)
2. Identify and analyze the requirements and construct their models (K5)
3. Apply systematic procedure for software design (K3)
4. Compare and contrast the various testing and maintenance activities (K2)
5. Make use of agile development and Devops (K3)
6. Identify unethical issues and apply ethical practices for a given case study (K3)

TEXTBOOKS

1. Roger S Pressman, Bruce R Maxin “Software Engineering – A Practitioner’s Approach”, McGraw Hill International Edition, Eighth Edition, 2015.
2. Armando Fox and David Patterson, Engineering Software as a Service: An Agile Approach Using Cloud Computing”, Strawberry Canyon LLC, Second Beta Edition, 2021.

REFERENCE BOOKS

1. Ian Sommerville, “Software Engineering”, Pearson Education Asia, Tenth Edition, 2015.
2. Stephen R Schach, “Software Engineering”, Tata McGraw-Hill Publishing Company Limited, 2007.
3. Brian Albee, Hands-On Software Engineering with Python, Packt Publishing, 2018.
4. Kelkar S A, “Software Engineering”, Prentice Hall of India, 2007.

COURSE PLAN

COURSE OUTCOMES (CO)

Students who successfully complete this course should be able to:

- CO1- Choose an appropriate process model and estimate project cost and effort required by applying software engineering principles(K5)
- CO2- Identify and analyze the requirements and construct their models (K5)
- CO3- Apply systematic procedure for software design (K3)
- CO4- Compare and contrast the various testing and maintenance activities (K2)
- CO5- Make use of agile development and Devops (K3)
- CO6- Identify unethical issues and apply ethical practices for a given case study (K3)

Content Delivery Methods (CDM): P: Presentations, D: Demo, PV: Pre -class Video, IV – In-class video, PS – Problem Solving, GD – Group Discussions, Quiz

Delivery Tools: PowerPoint presentations\ Whiteboard applications\ Projector\ Board\LMS

Assessment Tools : Test, Quiz, Tutorial, Assignments, Seminar, Team Project.

Topic		K level	CDM	COs	No of Sessions		Deviations if any
					Proposed	Actual	
UNIT – I [SOFTWARE PROCESS AND PLANNING] CO1 – K5					9		
	Introduction to Software Engineering; Objectives, Principles and Practices	K2	P	CO1	1		
2	The Software Development Life Cycle : Pre-development phases of the SDLC – Development specific phases of the SDLC – Post-development phases of the SDLC	K5	P	CO1	1		
3	Process methodologies	K5	P	CO1	1		
4	Development paradigms – Development practices	K2	P	CO1	1		
5	Project Planning Process	K3	P	CO1	1		
6	Software Project Estimation: Decomposition techniques	K3	P	CO1	1		
7	Empirical estimation models – The make/buy decision	K3	P	CO1	1		
8	Project scheduling	K3	PS	CO1	1		
9	Risk Management & Handling Ethical Dilemmas	K3			1		
UNIT – II [REQUIREMENTS ANALYSIS AND SPECIFICATION] CO2 – K5					9		
10	Software Requirements: Functional and non-functional, Security requirements	K3	PS	CO2	1		
11	User requirements, System requirements	K3	P	CO2	1		
12	Software requirements document	K5	P	CO2	1		
13	Requirement Engineering Process: Feasibility studies	K2	P	CO2	1		
14	Requirements elicitation and analysis	K3	P	CO2	1		
15	Requirements validation	K3	P	CO2	1		
16	Requirements Management	K2	P	CO2	1		
17	Classical Analysis: Structured system analysis	K2	P	CO2	1		
18	Requirement modelling tools	K3	P	CO2	1		
UNIT – III [SOFTWARE DESIGN] CO3 K3					10		
19	Design Concepts: Design process	K2	P	CO3	1		
20	Design concepts – Modularity	K3	P	CO3	2		
21	Coupling and cohesion	K3	P	CO3	1		
22	Design model, Modeling principles	K2	P	CO3	1		
23	Structured Design	K2	P	CO3	1		
24	Architectural Design: Architectural styles	K3	P	CO3	1		
25	Architecture for Network based Applications	K3	P, GD	CO3	1		
26	Decentralized Architectures.	K3	P	CO3	1		
27	Content Beyond Syllabus : IoT Architecture design	K2	P	CO3	1		
UNIT – IV [SOFTWARE TESTING] CO4 – K3					9		
28	Software Testing Fundamentals	K2	P	CO4	1		
29	Internal and External Views of Testing	K2	P	CO4			
30	White box testing, Basis path testing	K3	P	CO4	1		
31	Control structure testing, Black box testing	K3	P, Quiz	CO4	1		
32	Unit testing & Integration testing	K3	P	CO4	1		
33	Regression testing & Validation testing	K2	P	CO4	1		
34	System testing, Security testing	K3	P	CO4	1		

35	Testing Tool & Debugging	K3	P	CO4	1		
36	Software Implementation: Coding Practices and Principles	K2	P	CO4	1		
37	Maintenance: Types	K2	P	CO4	1		
UNIT – V [AGILE DEVELOPMENT AND DEVOPS] CO5 – K3					9		
38	Agile Development: Agile Teams	K2	P	CO5	1		
39	Team and Scrum – Branches	K3	P, Quiz	CO5			
40	Pull Requests – Reviews	K3	P	CO5	1		
41	Integration – Agile Iterations	K2	P	CO5	1		
42	Reporting and fixing bugs	K3	P	CO5	1		
43	Dev/Ops: From development to deployment – Three-Tier – Responsiveness	K2	P	CO5	1		
44	Service level objectives, and Apdex – Releases and feature flags	K2	P	CO5	1		
45	Monitoring and finding bottlenecks	K3	P	CO5	1		
46	Improving rendering and database performance with caching	K3	P	CO5	1		
47	Security: Defending customer data in application.	K2	P	CO5	1		

Reasoning for Content Beyond Syllabus: In Unit III, the current architecture design related to IoT will be discussed. The students know about the current trend in IoT architecture.

Total number of hours in the Syllabus: 45

Total number of hours planned: 46

Mapping of Course Outcomes with Program Outcomes

O/PSO mapping using Performance Indicators

O /PSO	CO1		CO2		CO3		CO4		CO5		CO6	
	PI	MS	PI	MS	PI	PI	PI	MS	PI	MS	PI	MS
1	1.3.1(3)	3	1.3.1(3) 1.4.1(3)	3	1.3.1(3)	3	1.3.1(3)	3	1.3.1(3)	3	1.3.1(3)	3
2	2.1.2(3)	2	2.1.2(3) 2.3.2(2)	3	2.2.1(2)	2	2.4.2(2)	2	-	-	-	-
3	-	-	3.1.2(2) 3.2.2(3) 3.2.3(2)	3	3.4.1(2)	2	3.4.3(3)	2	-	-	-	-
4	-	-	-	-	-	-	-	-	-	-	-	-
5	-	-	5.1.2(3)	3	-	-	-	-	-	-	-	-
6	-	-	6.2.1	2	-	-	-	-	-	-	-	-
7	-	-	7.1.1	3	-	-	-	-	-	-	-	-
8	-	-	-	-	-	-	-	-	-	-	8.1.1(3) 8.2.2(3)	3
9	-	-	9.1.2(3)	3	-	-	-	-	-	-	-	-
10	-	-	10.1.2(3)	3	-	-	-	-	-	-	10.1.2(3)	3
11	-	-	-	-	-	-	-	-	-	-	-	-
12	-	-	-	-	-	-	-	-	-	-	12.1.2(3)	3
PSO1	-	-	13.3.1(2) 13.3.2(2)	2	13.3.2(2)	2	13.4.2(3)	3	-	-	-	-
PSO2	14.1.1(3)	3	14.1.1(3) 14.2.2(2)	3	14.1.2(2)	2	14.2.1(3)	3	-	-	-	-

PO/ PO and PIs:

1.3.1	Apply engineering fundamentals	3
1.4.1	Apply theory and principles of computer science and engineering to solve engineering problems	3

2.1.2	Identify processes/modules/algorithms of computer-based systems and parameters to solve problems	2
2.2.1	Reframe computer-based systems into interconnected subsystems	2
2.3.2	Identify design constraints for required performance criteria	2
2.4.2	Analyze and interpret results using contemporary tools	2
3.1.2	Identify and document system requirements	2
3.2.2	Produce a variety of potential design solutions suited to meet functional requirements	3
3.2.3	Identify suitable non-functional requirements for evaluation of alternative design solutions	2
3.4.1	Refine architecture design into a detailed design within the existing constraints	2
3.4.3	Verify functionalities and validate the design	3
5.1.2	Create/adapt/modify/extend tools and techniques to solve engineering problems	3
6.2.1	Interpret legislation, regulations, codes, and standards relevant to computer science and engineering and explain its contribution to the protection of the public	2
7.1.1	Identify societal and environmental risks in engineering solutions	3
8.1.1	Identify situations of unethical professional conduct and propose ethical alternatives	3
8.2.2	Examine and apply moral and ethical principles to known case studies	3
9.1.2	Implement the norms of practice (rules, roles, charters, agendas, etc.) of effective teamwork, to accomplish a goal	3

	produce clear, well-constructed, and well-supported written engineering documents	3
	Describe the rationale for continuous professional development	1
2.1.2	Identify deficiencies or gaps in knowledge and demonstrate ability to find information to close these gaps	3
13.3.1	Apply fundamentals of problem solving	2
13.3.2	Design solutions in a modular way	2
13.4.2	Apply tools and validation methods to evaluate the solutions	3
14.1.1	Conduct feasibility analysis and cost-benefit analysis	3
14.1.2	Apply different design methods in software development	2
14.2.1	Verify the correctness of systems	3
14.2.2	Apply best practices to improve the quality of development and testing activities	2

COURSE OUTCOMES (CO) MAPPED TO PROGRAMME OUTCOMES (PO) AND PROGRAM SPECIFIC OUTCOMES (PSO)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2												3
CO2	3	3	3		3	2	3		3	3	3		2	3
CO3	3	2	2										2	2
CO4	3	2	2										3	3
CO5	3													
CO6	3							3		3		3		
Score	18	9	7		3	2	3	3	3	6	3	3	7	11
Course Mapping	3	3	3		3	2	3	3	3	3	3	3	3	3

3 – Strong 2 – Moderate 1 – Reasonable

Assessment Tools for Assessing COs

How course outcomes will be assessed:

Assessment Tool	Unit	CO1	CO2	CO3	CO4	CO5	CO6
CAT-1	I, II & III	Y	Y	Y			
CAT-2	III, IV & V			Y	Y	Y	
Assignment KL (K5)	I & II	Y	Y				Y
Assignment KL (K5)	III & IV			Y	Y		Y

CAT 1 assesses CO1, CO2 and CO3 (from Units I, II and III) mapping to PO1, PO2, PO3, PO5, PO6, PO7, PO9, PO10, PO11, PSO1, and PSO2.

CAT 2 assesses CO3, CO4 and CO5 (from Units III, IV and V) mapping to PO1 PO2, PO3, PSO1, and PSO2.

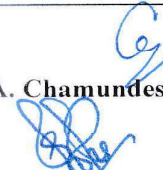
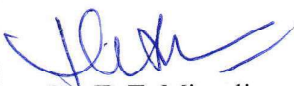
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Assessment Format and Details

Assessment				
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- Assignments with a weightage of 10% as per the course instructor's schedule.
- End Semester Examination with a weightage of 60% as per final examination schedule.

Prepared By	Verified by	Approved By
 Dr. A. Chamundeswari  Dr. Y. V. Lokeswari (Course In-charges)	 UG-PAC Team	 Dr. T. T. Mirnalinee (HOD - CSE)